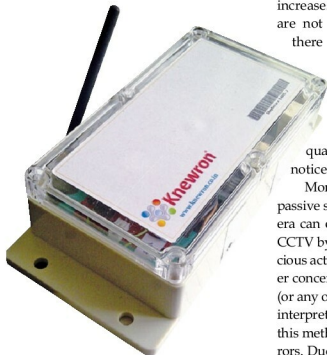


ShopSecure: An IoT Security System

ShopSecure is an active security monitoring system developed by Knewron, which not only detects security breaches and unusual activities but also informs the occurrence to the user and authorities



ShopSecure security monitoring system

ANAGHA P.

We now live in a world where we are being watched everywhere. Seems like there are surveillance cameras in all places you go. You may not be aware of it but almost all retail shops, restaurants, ATM counters, banks and government establishments are equipped with surveillance cameras as a security measure. A person is likely to be caught in a security camera at least once a day.

Why not closed-circuit television (CCTV)

The closed-circuit television (CCTV) cameras are very popular amongst the shop owners and security specialists but they have several shortcomings.

First of all, the CCTV cameras are used for live monitoring during the daytime, but when a shop is closed, the cameras are also disabled, and this

increases vulnerability. Even if they are not disabled after closed hours, there is no way for the employer to know about a security breach until the footage for previous night is checked. Sometimes when only a few goods are stolen from a large quantity of stocks, it may go unnoticed until reconciliation.

Moreover, CCTV monitoring is a passive security system where the camera can only record the video, but the CCTV by itself cannot detect any suspicious activity or inform it to whomsoever concerned. Since you need a human (or any other specialised mechanism) to interpret the feed and act accordingly, this method is tedious and prone to errors. Due to all these factors, the CCTV method is almost meaningless when it comes to intrusion detection.

The birth of ShopSecure

The distribution partners of Knewron, S.V. TechnoCrafts, were approached by shop owners in their area who said that many of them were desperately looking for something which could serve as an effective alternative to CCTV cameras and be economical for their pockets. "This critical need prompted us to work on a solution that could serve the purpose with the help of technology at hand," says T. Anand, the co-founder of Knewron.

Imagine a device that is user-friendly, economical, reliable, records the video round the clock, does some analysis by itself, detects suspicious activities and takes precautionary measures. You need an active system for fulfilling these conditions and ShopSecure is one such system. It can detect malfunctioning or suspicious incidents and report or act accordingly.

ShopSecure would not wait until the footage is analysed by the user, but it will intimate them instantly of security breach and can also sound an alarm to scare away intruders.

Working

ShopSecure belongs to the Internet of Things (IoT) device line with fundamental IoT qualities. Basically it is an M2M communication model which is interoperable in nature. It involves uniquely addressable physical entity connected in local/private or cloud network for effective functioning. The device activity log can be streamed over the Internet and later analysed by users. The device can operate autonomously with Internet Bridge over cloud (with GPRS or 3G) or in direct contact mode (with GSM module).

All the intelligence and processing of ShopSecure comes from Atmel's 8-bit AVR microcontroller core around which this device is built. Temperature sensor, audio playback and battery chargeover recharging unit are the other key components of this device.

The device is equipped with infiltration sensors (MEMs, reed switches or micro-switches, etc) which actively monitor environment and doors/windows when armed. It is designed to intimate the user when it senses anything out of the ordinary, like a sudden rise in temperature, and it can take appropriate predefined actions. ShopSecure can be remotely controlled and configured and is constructed from some of the commonly available sensors and other components, thus making it low-cost. Its configuration and operation are so simple that, if you can use a mobile phone, you can use it.

In case of devices deploying CCTVs or IR sensors and other proximity sen-